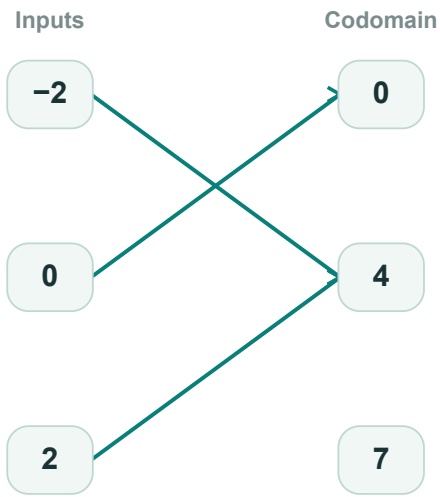
FG-A1 / One input, one output

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Trace a valid function

Tap −2, then 2. Follow the arrows.



f(−2)=4 and f(2)=4

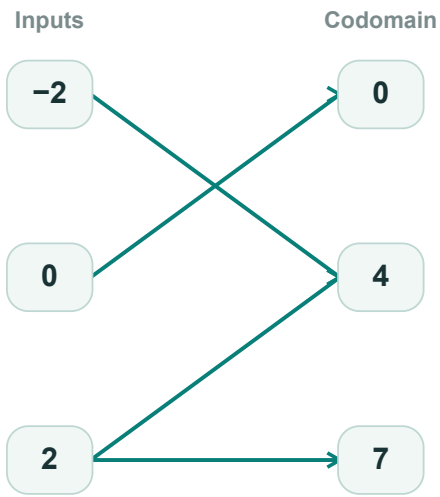
Tap an input

Different inputs may share an output.

Previous Next →

After checking an invalid relation

Input 2 now has two different outputs.



2 maps to 4 and to 7

Choose an arrow to remove

One permitted input needs exactly one output.

Previous Next →

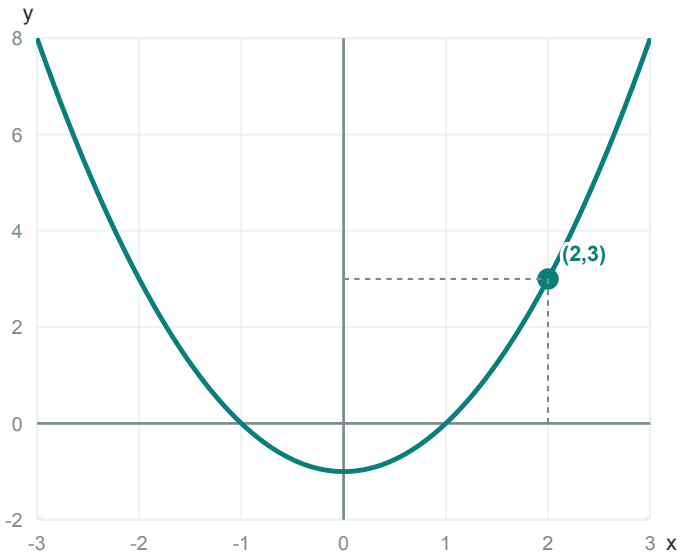
Mapping buttons provide the primary action; arrows explain the relationship.

FG-A2 / From a rule to a graph

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Read an output

Move the input marker to 2.



f(x)=x²−1; f(2)=3

Input x = 2

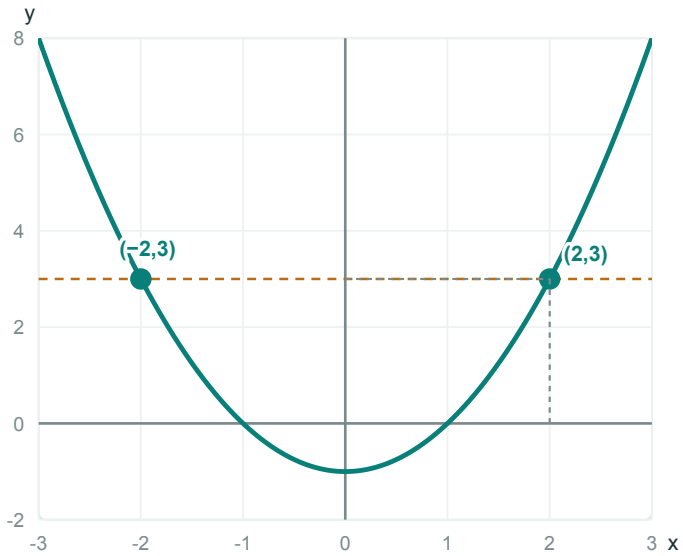
The point (2,3) records an input and its output.

Previous

Next →

Find the inputs

Set the output to 3. Select every input.



f(x)=3 at x=−2 and x=2

Output level = 3

A horizontal question can have more than one answer.

Previous

Next →

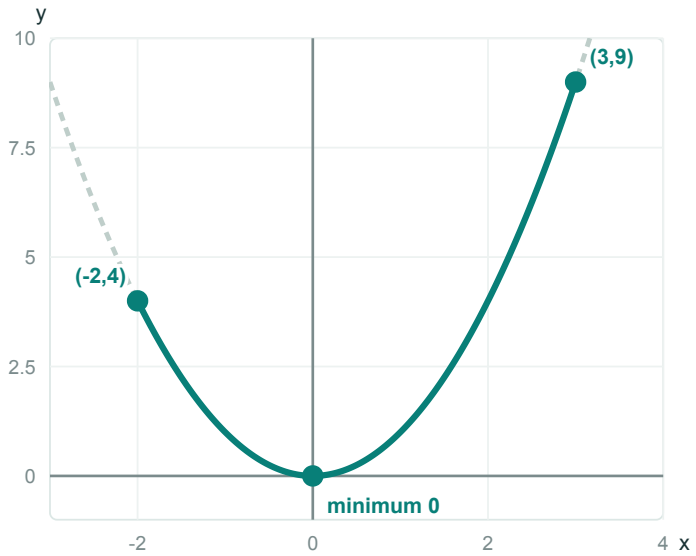
Linked point, table row and tracing guides make the representation change explicit.

FG-A3 / Domain, range and codomain

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Start with a restricted domain

Trace all allowed inputs.



Domain [−2,3]; range [0,9]

Left endpoint −2 · Right endpoint 3

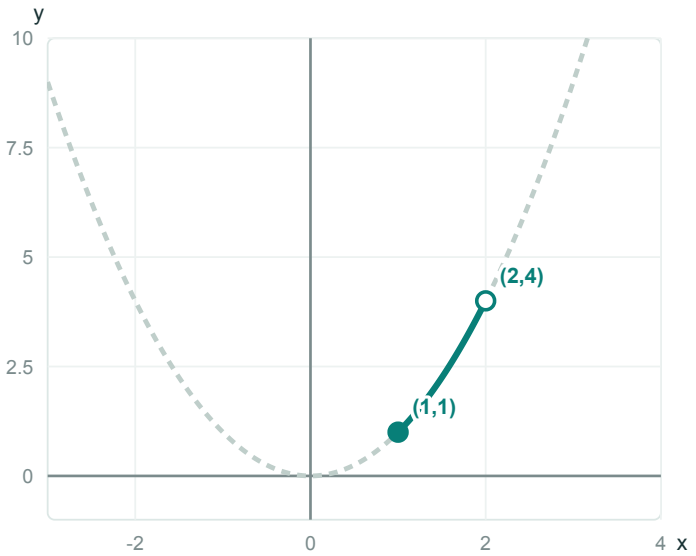
The minimum output lies inside the interval.

Previous

Next →

After changing the domain

Keep 1≤x<2. Compare the attained outputs.



Domain [1,2); range [1,4)

Right endpoint: excluded

The original formula is unchanged; permitted pairs changed.

Previous

Next →

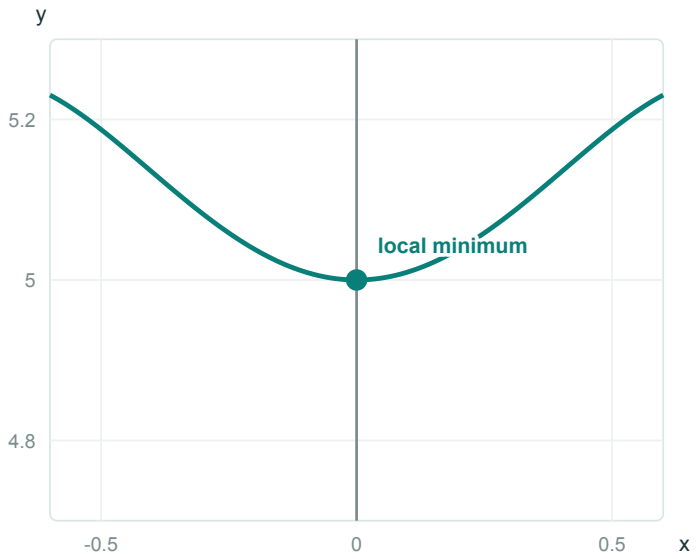
A dashed curve is context only: those inputs are excluded, not part of this function.

FG-A4 / Reading change, signs and extremes

ANNOTATED DESIGN SKETCH · selected states, not a production screen

A nearby view

Inspect the valley around x=0.



$h(x)=5+x^2-x^4$

Window: −0.6≤x≤0.6

There is a local minimum at (0,5).

Previous

Next →

The same function, wider view

Compare the valley with h(2).



$h(2)=-7 < 5$

Window: −2≤x≤2

Local minimum does not mean lowest everywhere.

Previous

Next →

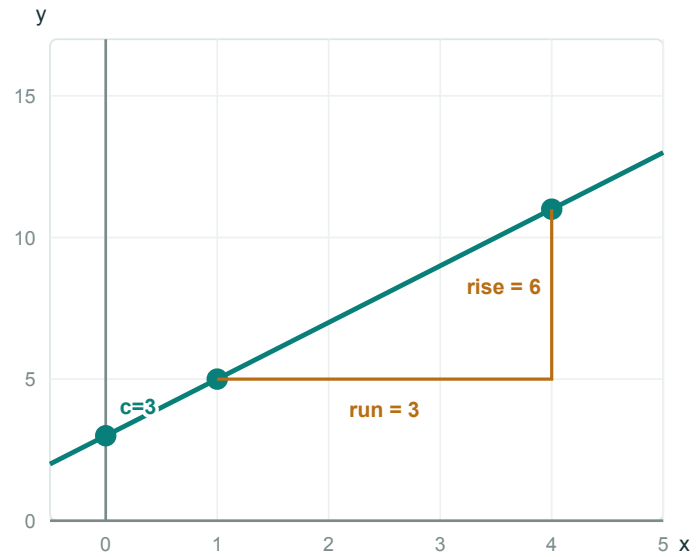
Only the viewing window changes. The declared domain remains all real inputs.

FG-A5 / Lines: a constant rate

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Measure the changes

Move the second point along the line.



y=2x+3; gradient 6/3=2

Second input x = 4

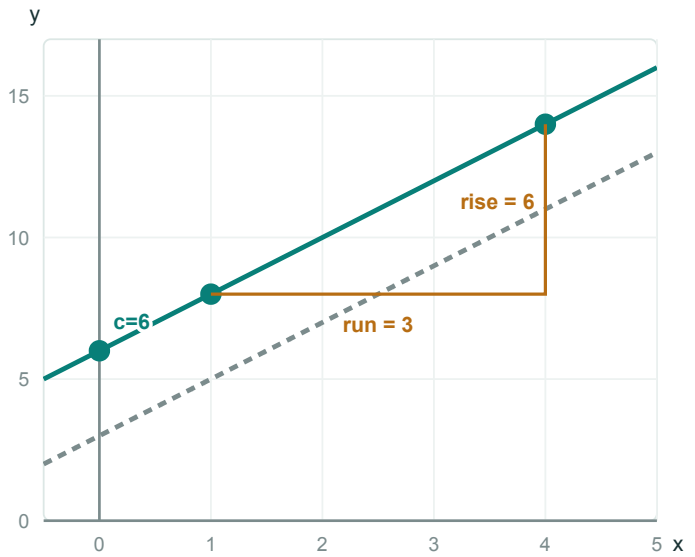
Rise and run vary together; their ratio stays 2.

Previous

Next →

Raise the starting amount

Change c from 3 to 6.



y=2x+6; gradient 6/3=2

Starting amount c = 6

A higher line need not have a higher rate.

Previous

Next →

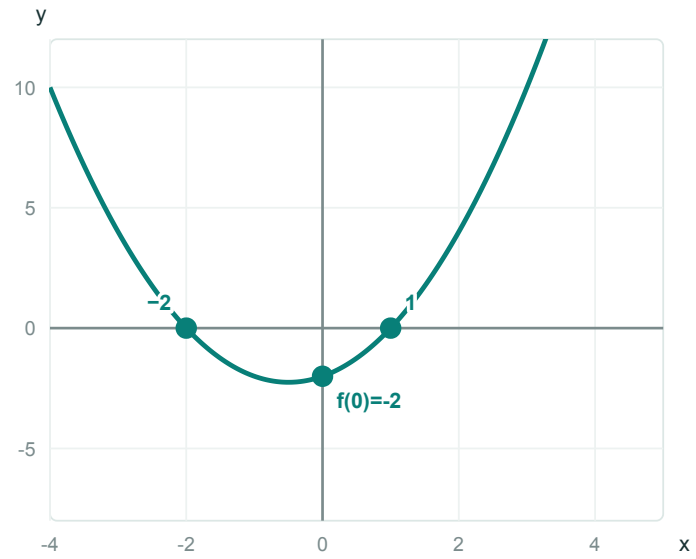
Dashed line preserves the original position; graph scales remain fixed.

FG-B1 / Roots, factors and intercepts

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Start with a root at 1

Move the right root along the axis.



f(x)=(x+2)(x−1)

Moving root r = 1

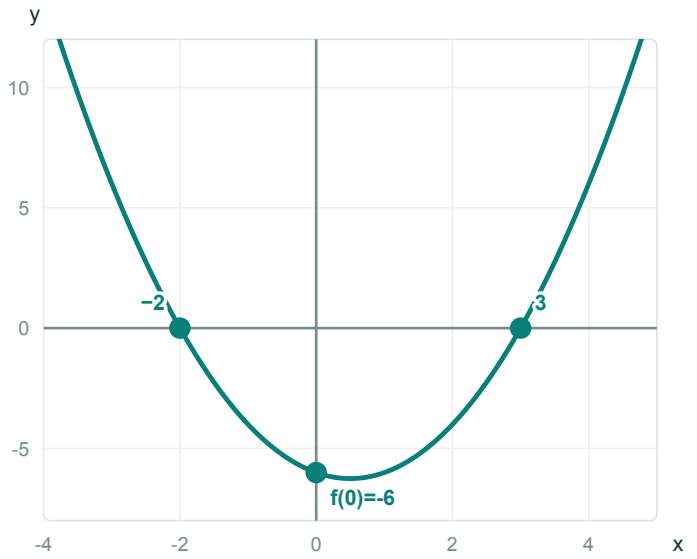
Substitution at either root makes a factor zero.

Previous

Next →

Move the root to 3

Match the root, factor and intercept.



f(x)=(x+2)(x−3)

Moving root r = 3

The fixed root stays −2 while f(0) changes.

Previous

Next →

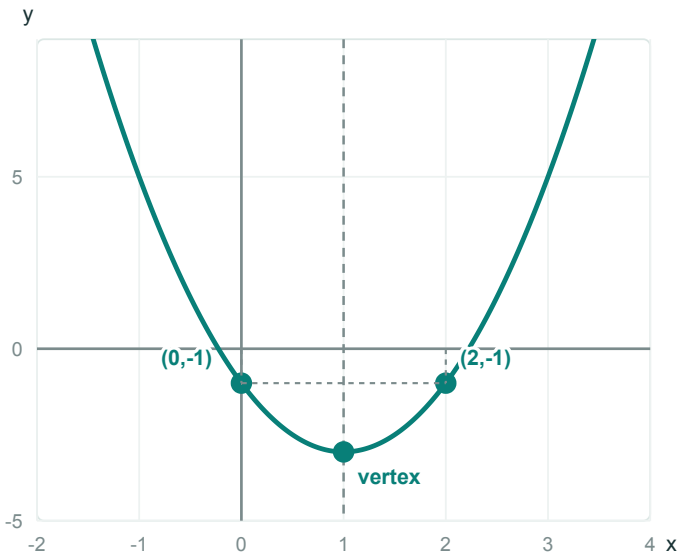
The existing roots interaction can be adapted; this plate adds the intercept connection.

FG-B2 / The vertex tells the shape

ANNOTATED DESIGN SKETCH · selected states, not a production screen

One unit from the centre

Move the paired points equally far from 1.



f(x)=2(x−1)²−3; d=1

Distance d = 1

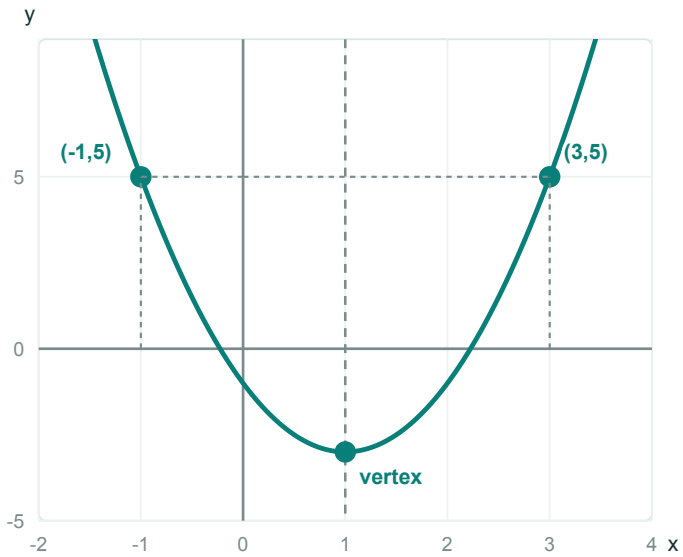
Both outputs are −1: two units above the vertex.

Previous

Next →

Twice the distance

Compare the heights above the same vertex.



f(x)=2(x−1)²−3; d=2

Distance d = 2

Both outputs are 5: eight units above the vertex.

Previous

Next →

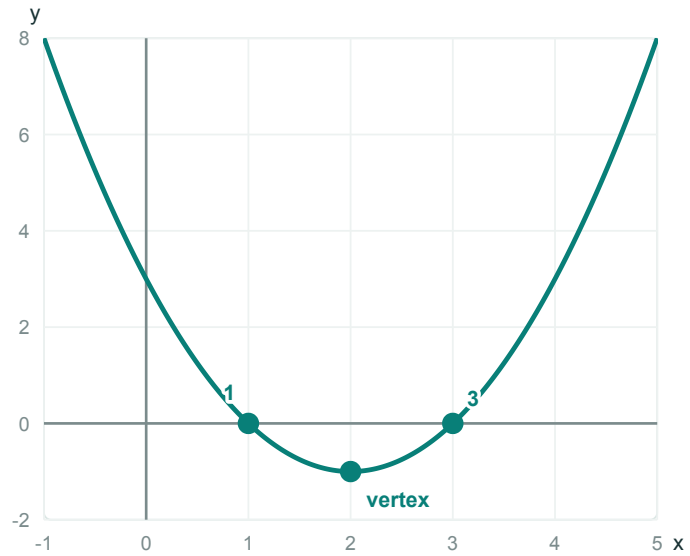
Equal distances produce equal heights; doubling distance quadruples the excess above k.

FG-B4 / When roots meet and disappear

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Two distinct real roots

Raise c towards the touching case.



f(x)=x²−4x+3; Δ=4

Parameter c = 3

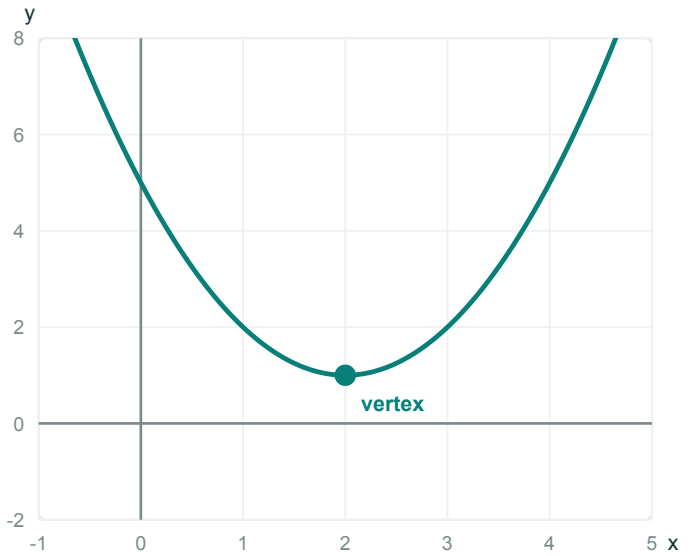
The real roots are 1 and 3.

Previous

Next →

Beyond the threshold

Continue above c=4.



f(x)=x²−4x+5; Δ=−4

Parameter c = 5

The graph has real outputs, but none equals zero.

Previous

Next →

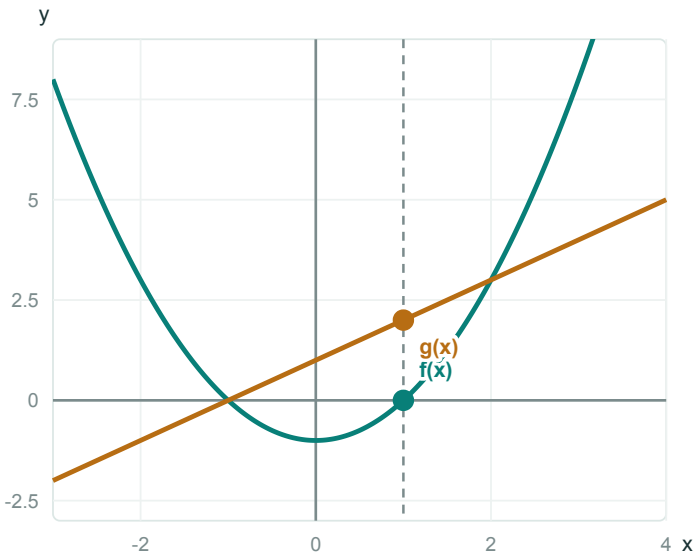
At c=4 the two roots coincide at x=2. These panels show either side of that event.

FG-C1 / Equal outputs: intersections

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Compare at the same input

Move x until f(x) and g(x) agree.



f(x)=x²−1; g(x)=x+1

Input x = 1

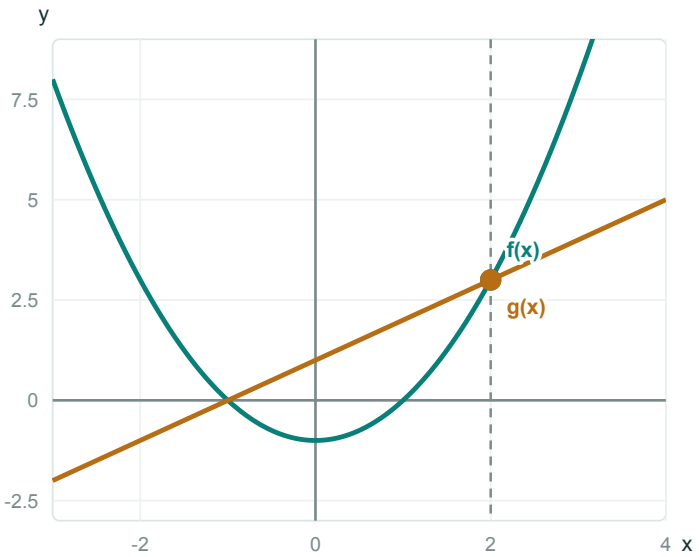
At x=1, the outputs are 0 and 2.

Previous

Next →

An intersection

Read both coordinates of the shared point.



f(2)=g(2)=3; f(2)−g(2)=0

Input x = 2

The input is 2; the intersection is (2,3).

Previous

Next →

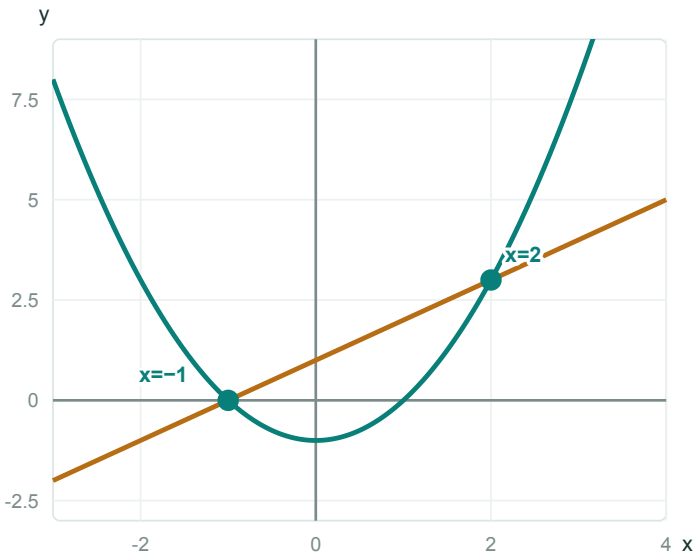
Teal is f; amber is g. A later stage turns their signed difference into a graph.

FG-C2 / Inequalities: whole intervals

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Make a comparison

Select every interval where f is above g.



f(x)=x²−1; g(x)=x+1

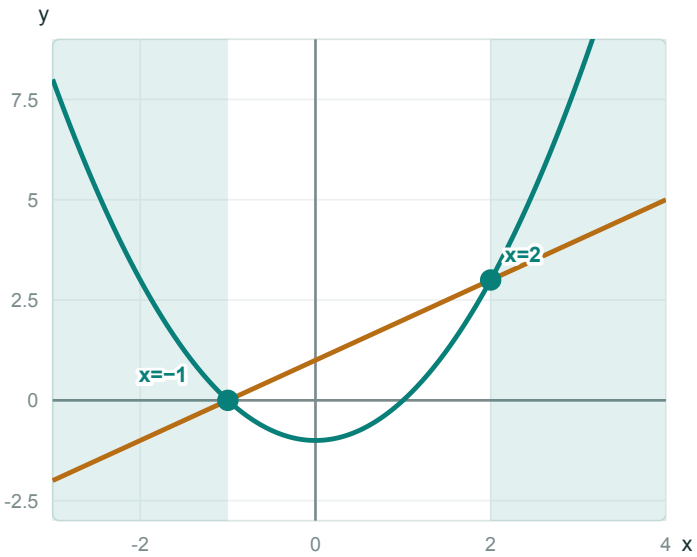
Choose intervals · Check

The intersections divide the input line.

Previous **Next →**

After checking the intervals

Read the complete solution set.



f(x)>g(x): x<−1 or x>2

Strict inequality: endpoints excluded

The solution is a set of inputs, not just the crossings.

Previous **Next →**

Bands show solution inputs; the actual intersection points still exist on the curves.

FG-C3 / Absolute value is distance

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Measure one distance

Move x around the centre at 2.



y=|x−2|; at x=0, distance=2

Position x = 0

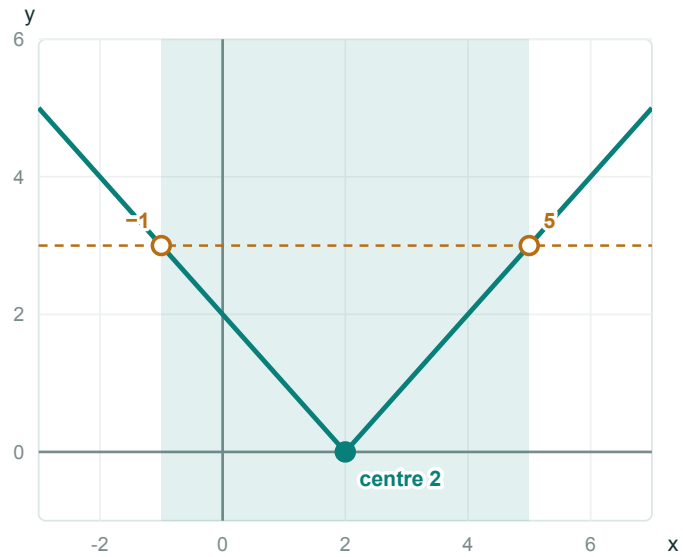
Distance is nonnegative on either side of 2.

Previous

Next →

Choose all close positions

Find distances less than 3.



|x−2|<3 gives −1<x<5

Distance threshold = 3

Between the boundaries, the V lies below y=3.

Previous

Next →

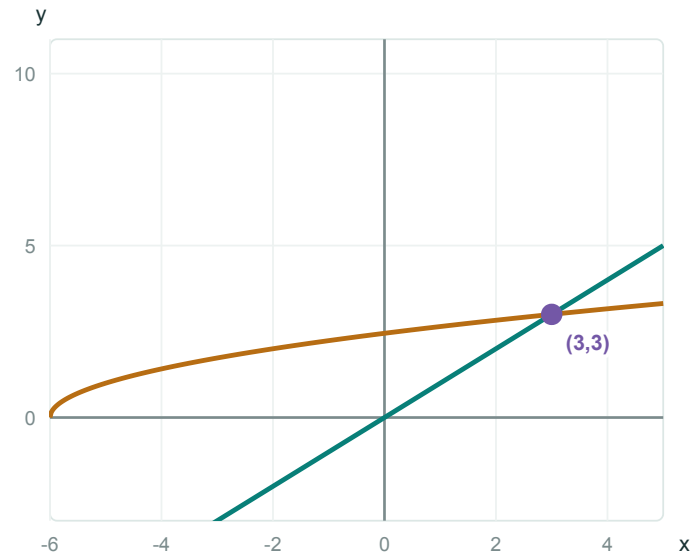
The final interaction also links this graph to a number line above it.

FG-C4 / Candidates are not yet solutions

ANNOTATED DESIGN SKETCH · selected states, not a production screen

The original equation

Find where the two outputs agree.



$x=\sqrt{(x+6)}$

View: original equation

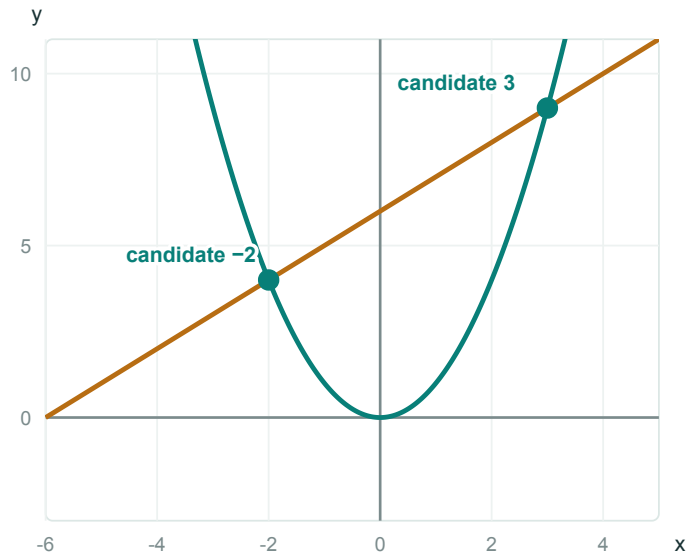
Only x=3 is a shared input.

Previous

Next →

After squaring

Test each candidate in the original equation.



$x^2=x+6$ → candidates −2 and 3

View: squared equation

At x=−2, the original sides are −2 and 2.

Previous

Next →

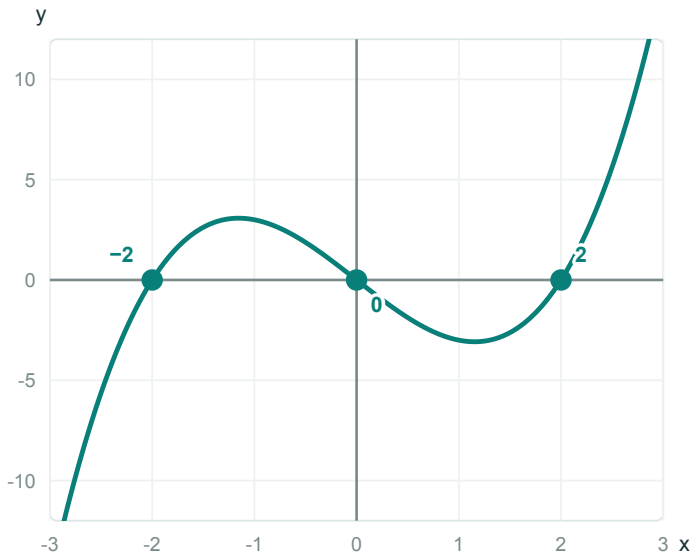
The second graph is a transformed equation. Its extra intersection is not an original solution.

FG-D1 / Cubics and the ends of a graph

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Three real intercepts

Inspect the roots and both ends.



f(x)=x³−4x

Choose a cubic preset

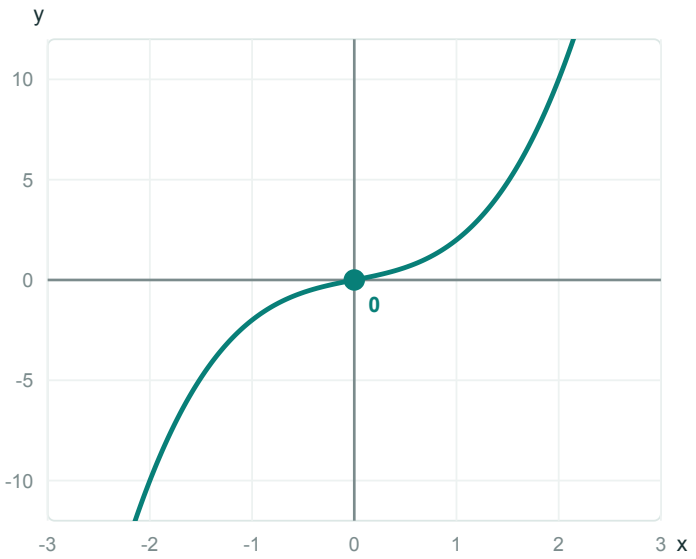
A cubic can have three distinct real roots.

Previous

Next →

One real intercept

Compare a different cubic.



g(x)=x³+x=x(x²+1)

Choose a cubic preset

Here x²+1 is always positive, so only x=0 is a root.

Previous

Next →

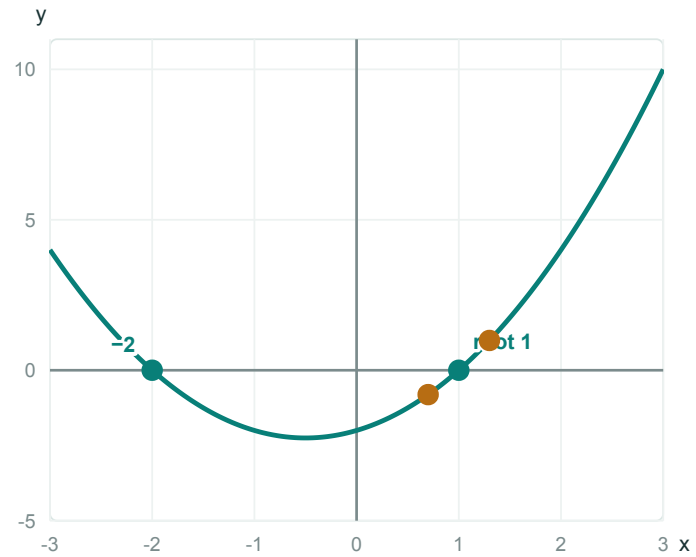
Degree and leading sign agree in both cases; the number of distinct real roots differs.

FG-D2 / Crossing, touching and repeated roots

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Power 1

Compare signs either side of x=1.



f(x)=(x−1)(x+2)

Exponent of (x−1): 1

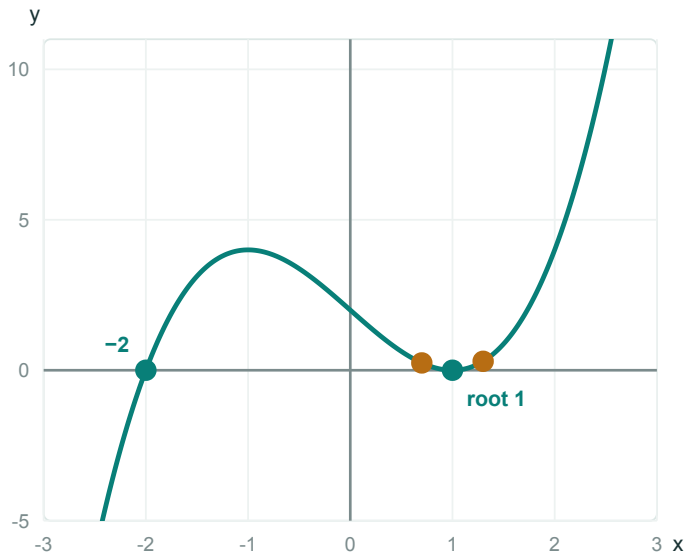
The sign changes across the simple root.

Previous

Next →

Power 2

Keep the root; change its multiplicity.



g(x)=(x−1)²(x+2)

Exponent of (x−1): 2

The graph touches at 1 and keeps its nearby sign.

Previous

Next →

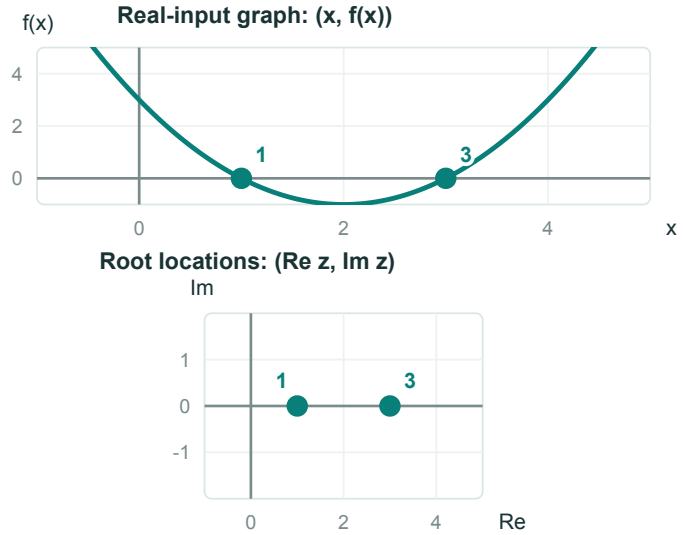
A third stage uses power 3 to reveal a flattened crossing. Other factors stay nonzero nearby.

FG-D6 / Where did the roots go?

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Two real roots

Compare the two kinds of diagram.



$(z-2)^2+k=0$; $k=-1$

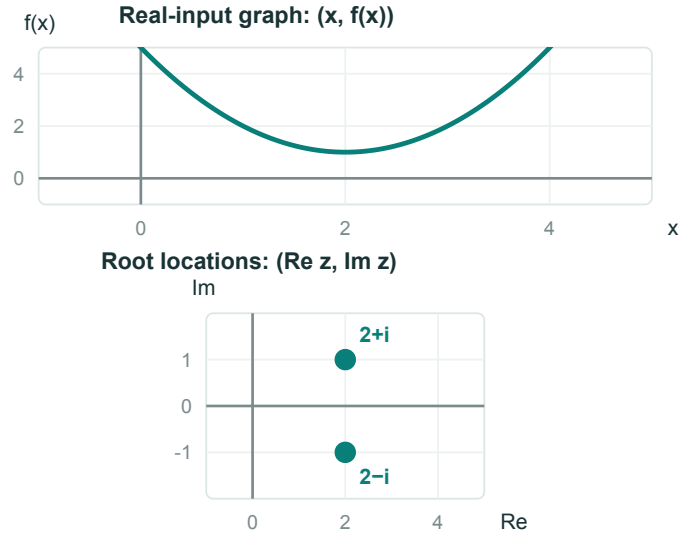
Parameter k = −1

Real roots 1 and 3 also lie on the Argand real axis.

Previous **Next →**

Two non-real roots

Increase k through zero to 1.



$(z-2)^2+1=0 \rightarrow z=2\pm i$

Parameter k = 1

No real intercepts; the root locations are 2+i and 2−i.

Previous **Next →**

The lower panel locates numbers. Its vertical coordinate is imaginary part, not f(x).

FG-E1 / Translations and their direction

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Predict the new point

For f(x+2), where does (1,3) move?



Original domain [−1,3]

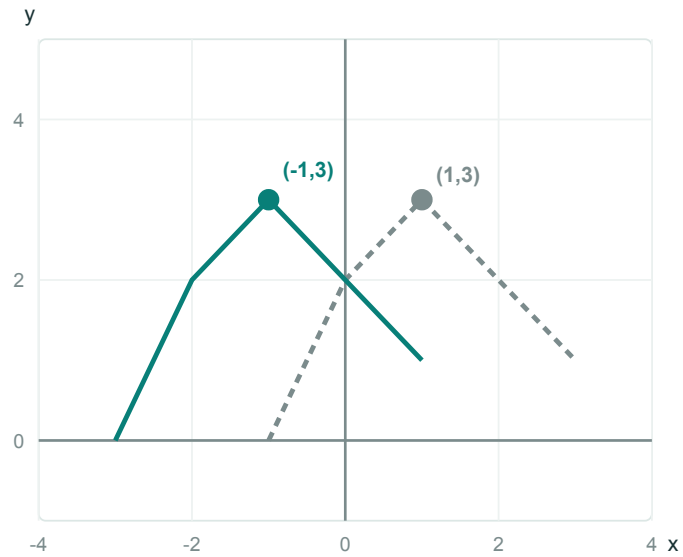
Inside amount +2 · Predict

Solve x+2=1 to reach the old input 1.

Previous Next →

After applying the translation

Follow the marked point and both endpoints.



New point (−1,3); domain [−3,1]

Show transformed graph

Plus inside reaches each old input two units earlier.

Previous Next →

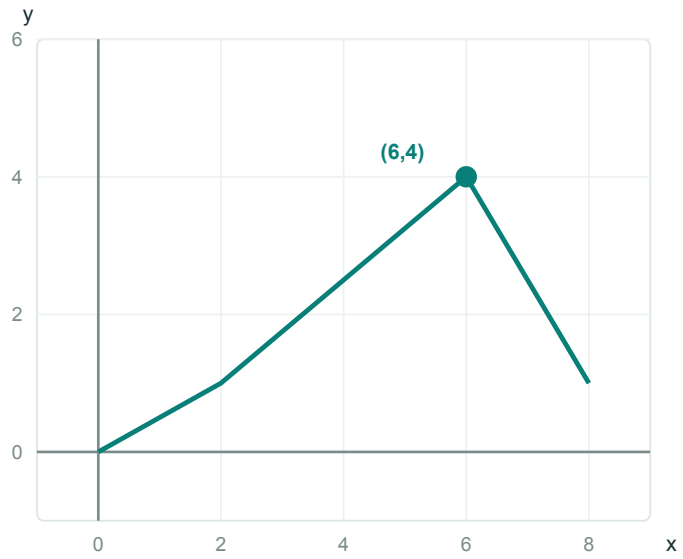
Original: dashed. Transformed: solid. All corresponding points obey the same rule.

FG-E2 / Reflections and scaling

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Predict a horizontal scale

For f(2x), find the new input for (6,4).



Old input = 6; inner input = 2x

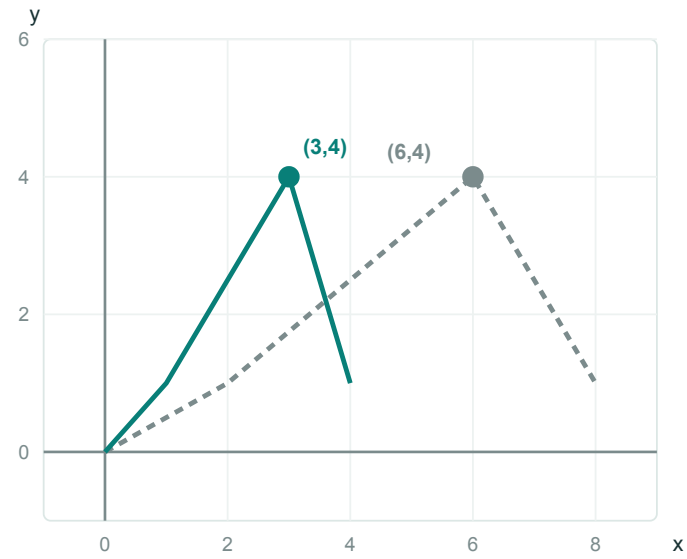
Input scale 2 · Predict

The output is unchanged; solve 2x=6.

Previous Next →

After applying the scale

Compare original and transformed widths.



(6,4) maps to (3,4)

Input scale = 2

Doubling inside halves the required horizontal coordinate.

Previous Next →

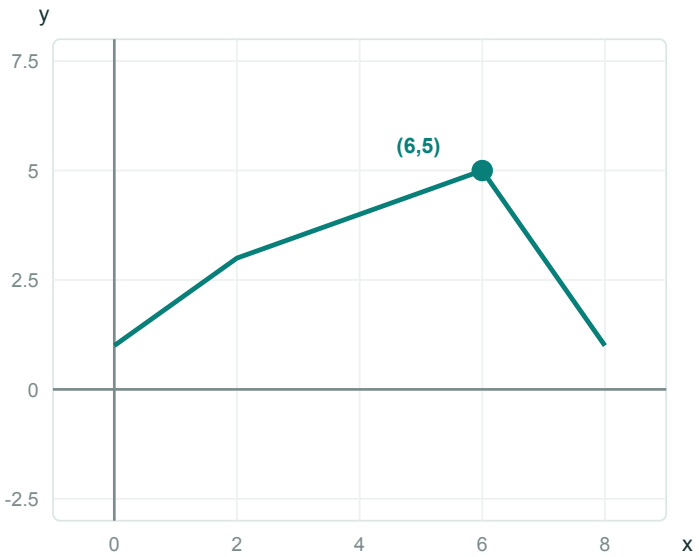
An asymmetric source graph keeps horizontal and vertical operations distinguishable.

FG-E3 / Combined transformations

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Track one point first

Apply g(x)=2f(3(x−1))−4.



Original point (6,5)

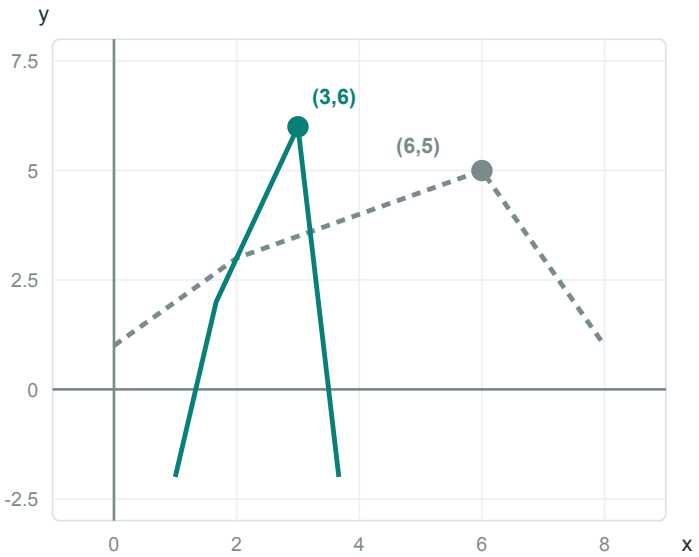
Predict corresponding point

Solve 3(x−1)=6, then calculate 2×5−4.

Previous Next →

Check the whole transformation

The same rule moves every landmark.



New point (3,6)

Show transformed graph

Input correspondence and output change settle the order.

Previous Next →

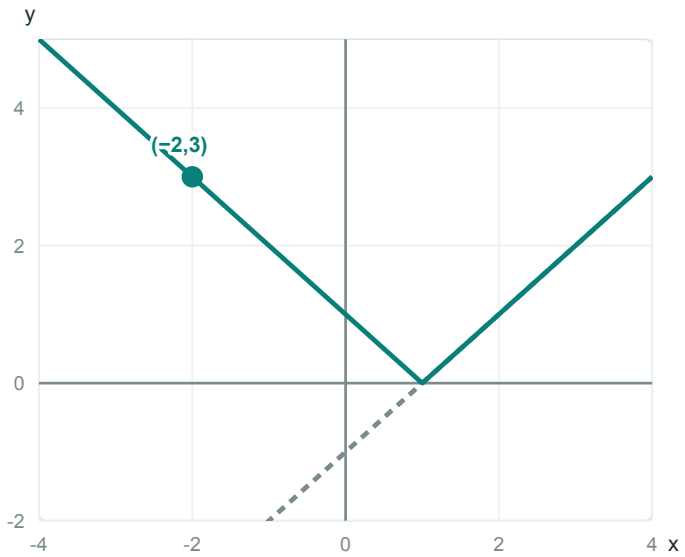
Domain endpoints use the same rule; [0,8] becomes [1,11/3].

FG-E4 / Two meanings of absolute value

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Change the output

Follow a negative output as it folds up.



|f(x)|=|x−1|

Mode: |f(x)|

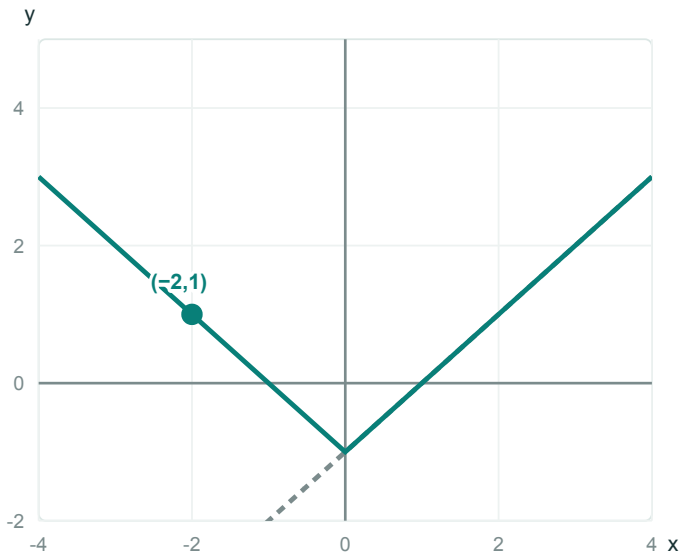
At x=−2, the new output is 3.

Previous

Next →

Change the input

Use the original positive-input branch twice.



f(|x|)=|x|−1

Mode: f(|x|)

At x=−2, the new output is 1.

Previous

Next →

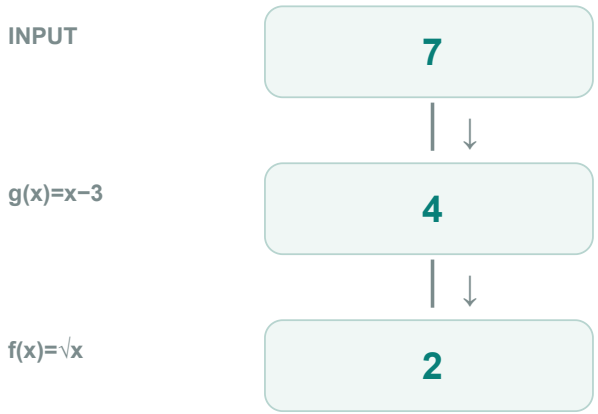
The dashed line is f(x)=x−1. Bar placement determines which quantity becomes nonnegative.

FG-F1 / Composition: output becomes input

ANNOTATED DESIGN SKETCH · selected states, not a production screen

A permitted input

Send 7 through g, then f.



f(g(x))=√(x−3)

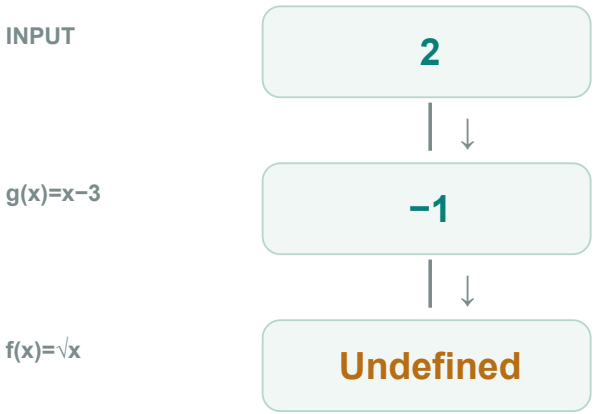
Input x = 7

The intermediate value 4 becomes the next input.

Previous Next →

An input that fails later

Try 2 in the same composition.



g(2)=−1; f cannot accept −1

Input x = 2

The first function works; the second rejects its output.

Previous Next →

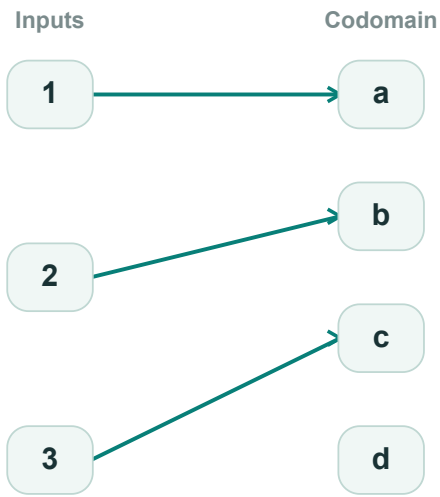
The intermediate value stays visible. Later graph mode uses the same relay without a formula.

FG-F2 / One-to-one, onto and bijective

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Look at the declared codomain

Which target is never reached?



Range {a,b,c}; codomain {a,b,c,d}

Select the unused target

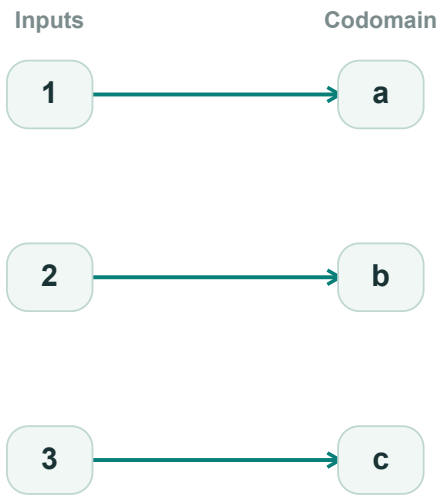
This mapping is injective but not onto.

Previous

Next →

Explicitly change the codomain

Remove only the unused target d.



Range = codomain = {a,b,c}

Codomain changed to {a,b,c}

The arrows did not move; onto status changed.

Previous

Next →

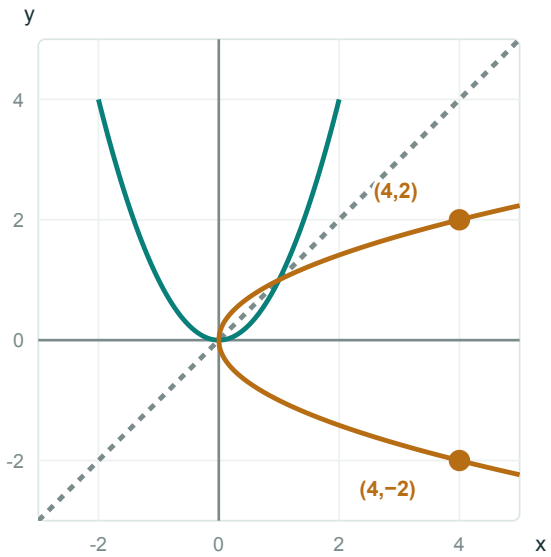
A valid codomain must still contain every actual output. Domain and codomain changes are labelled.

FG-F3 / Inverse graphs and a branch

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Reverse the full parabola

Inspect the reversed input 4.



Original $y=x^2$, $x \in \mathbb{R}$

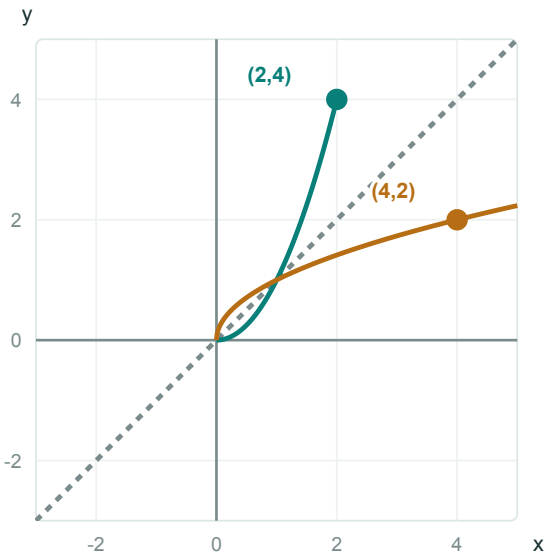
Swap coordinates

The inverse relation gives both 2 and −2 at input 4.

Previous **Next →**

Restrict the original domain

Keep $x \geq 0$ before reversing.



Inverse $y=\sqrt{x}$, $x \geq 0$

Original domain: [0,∞)

Now the reversed mapping has one output per input.

Previous **Next →**

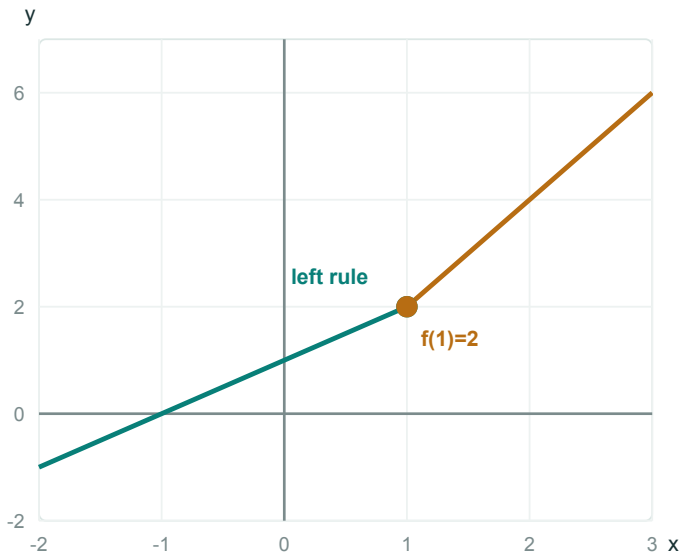
Square axis units preserve the reflection in y=x. Teal: original; amber: reversed pairs.

FG-G1 / Piecewise functions and boundaries

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Two rules that meet

Set x exactly to 1.



x+1 for x<1; 2x for x≥1

Input x = 1

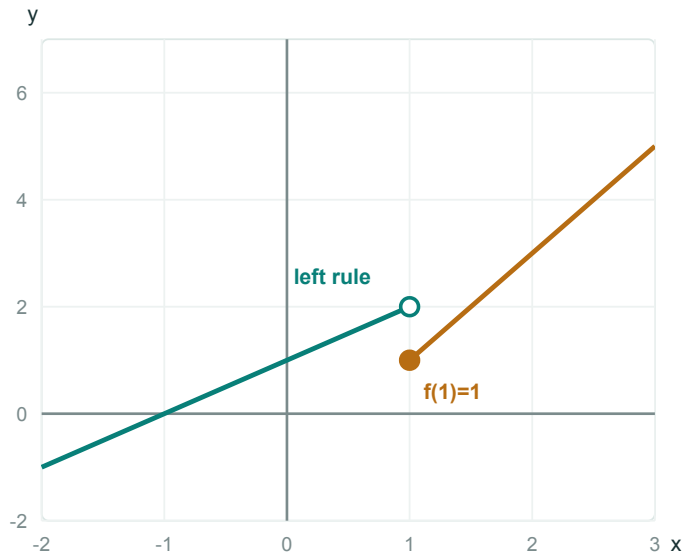
The second rule supplies f(1)=2.

Previous

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A different second rule

Compare the two boundary heights.



x+1 for x<1; 2x−1 for x≥1

Input x = 1

The open point is not f(1); the closed point gives 1.

Previous

Next →

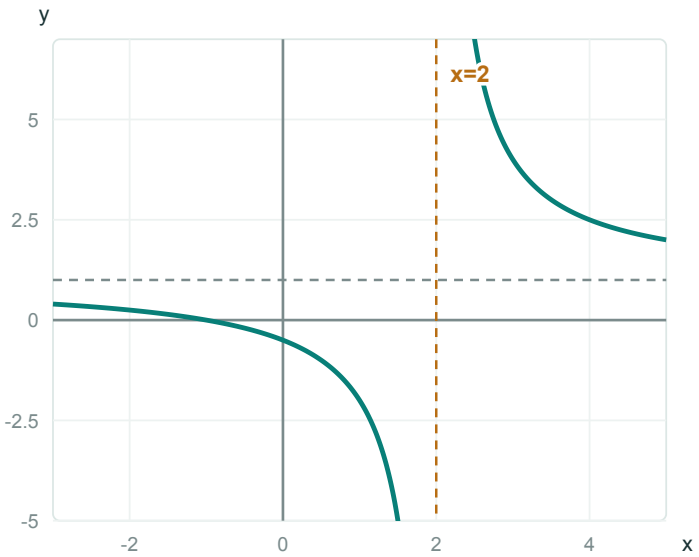
The condition selects the rule before calculation. No vertical connector joins a jump.

FG-G2 / Rational graphs: holes and asymptotes

ANNOTATED DESIGN SKETCH · selected states, not a production screen

An uncancelled denominator

Approach x=2 from either side.



f(x)=(x+1)/(x−2), x≠2

Choose: pole example

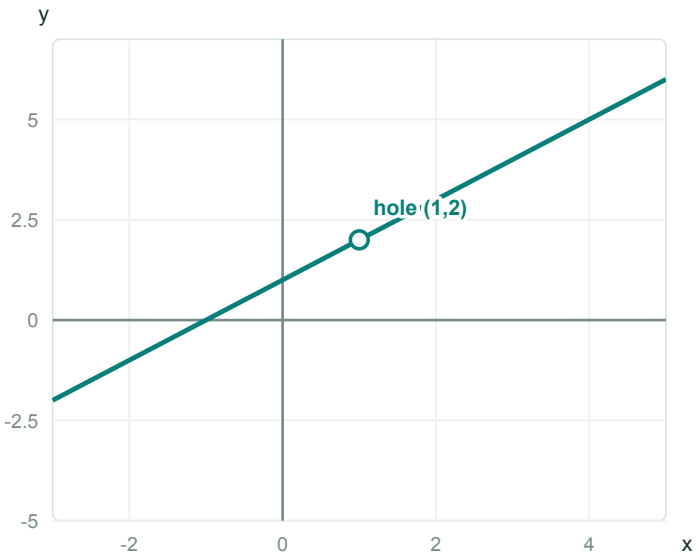
Outputs grow without bound near the excluded input.

Previous

Next →

A cancelled factor

Approach x=1; keep the original restriction.



g(x)=(x²−1)/(x−1), x≠1

Choose: hole example

Nearby outputs approach 2, but g(1) is undefined.

Previous

Next →

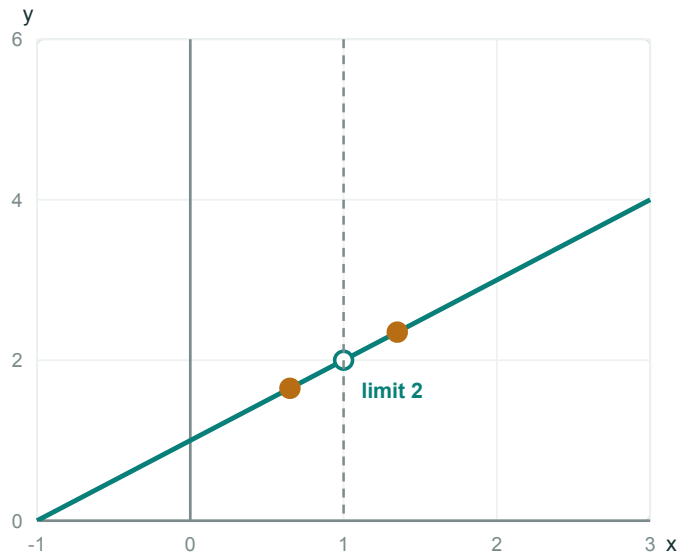
These are two different functions. Cancellation preserves the original missing input.

FG-G3 / The value there and the limit nearby

ANNOTATED DESIGN SKETCH · selected states, not a production screen

A missing value

Move the probes closer to x=1.



f(x)=x+1 for x≠1

Approach from both sides

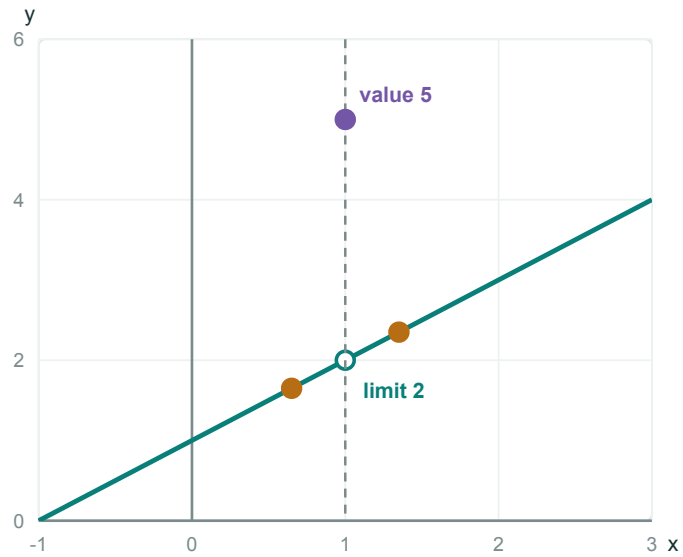
The limit is 2; there is no value at 1.

Previous

Next →

Assign one separate value

Set f(1)=5. Did the limit change?



Same nearby rule; f(1)=5

Point value = 5

The value changed. The neighbouring outputs did not.

Previous

Next →

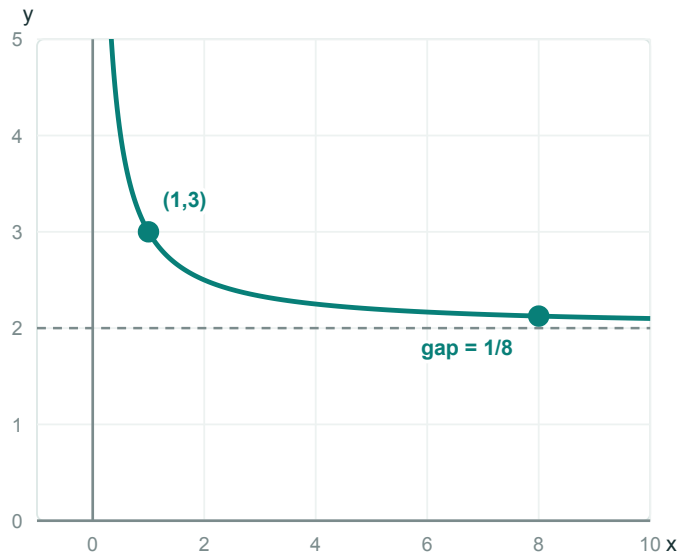
Setting f(1)=2 would fill the hole continuously; a later jump case has unequal one-sided limits.

FG-G4 / Long-run behaviour and asymptotes

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Approach a level

Choose larger positive inputs.



f(x)=2+1/x, x>0

Input x = 8 · Gap = 0.125

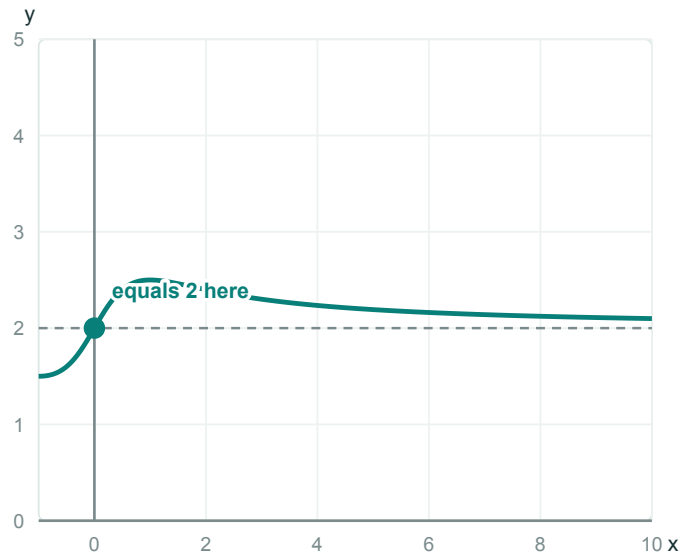
The gap 1/x tends to zero without being zero here.

Previous

Next →

An asymptote can be crossed

Check the output at x=0.



g(x)=2+x/(x²+1)

Compare: crossing example

g(0)=2, and g(x) still tends to 2 at both ends.

Previous

Next →

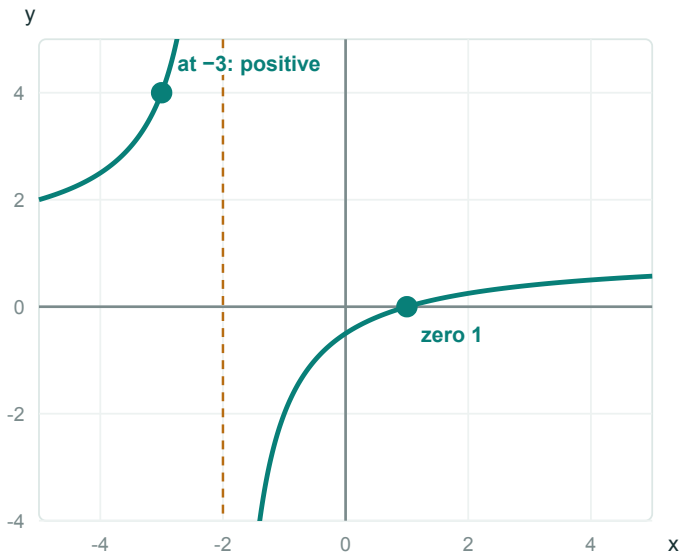
A horizontal asymptote describes long-run behaviour; it is not a forbidden output value.

FG-G5 / Rational inequalities and signs

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Test the original inequality

Check (x−1)/(x+2)≥0 at x=−3.



At −3: (−4)/(−1)=4 ≥ 0

Test input x = −3

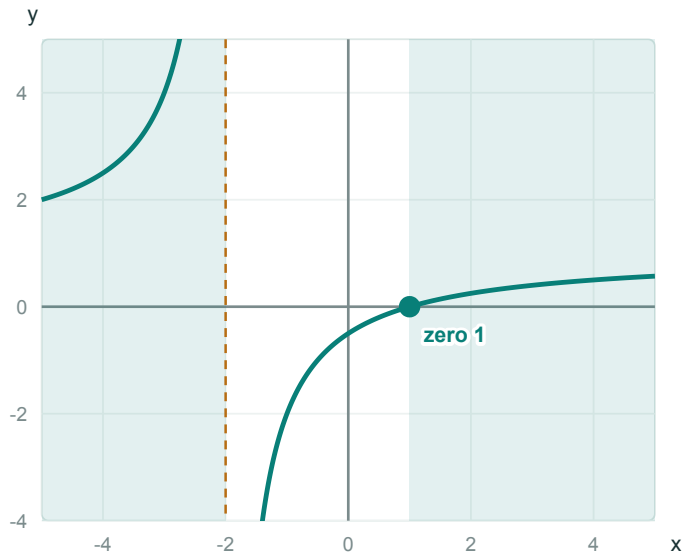
The negative denominator changes how multiplication behaves.

Previous

Next →

After sign analysis

Select the full solution set.



x<−2 or x≥1

Intervals selected · Check

Exclude −2; include the zero at 1.

Previous

Next →

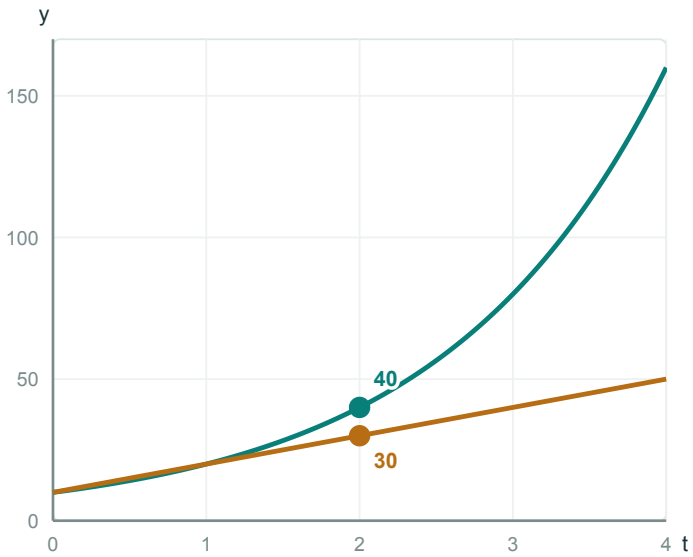
The pole and the root are different kinds of boundary. Shading denotes allowed solution inputs.

FG-H1 / Exponentials: equal ratios

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Compare equal time steps

Which model is larger after t=1?



$10 \cdot 2^t$ versus $10+10t$

Advance one time step

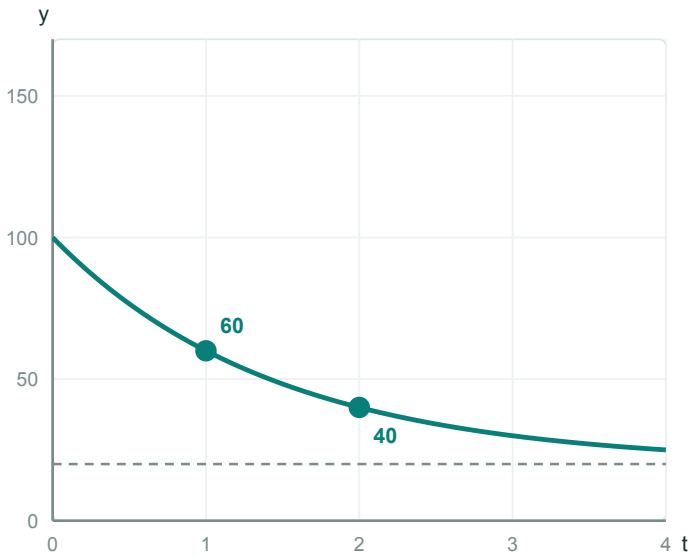
Same first two values; different later changes.

Previous

Next →

A baseline-shifted decay

Track the excess above 20.



$T(t)=20+80(1/2)^t$

Advance one time step

100 → 60 → 40: the excess halves, not T itself.

Previous

Next →

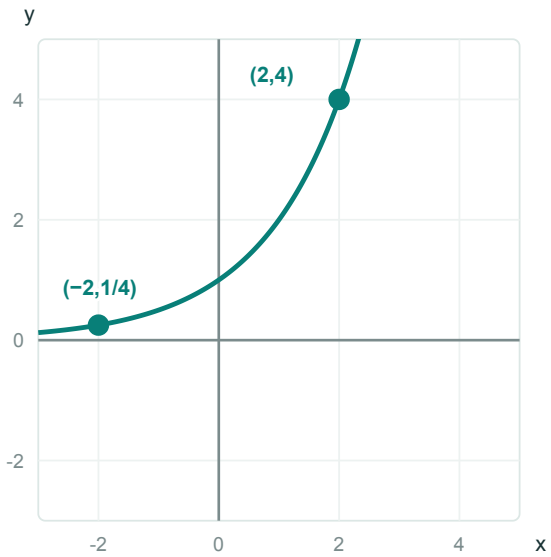
The comparison changes examples explicitly. Baseline and initial value are different quantities.

FG-H2 / Logarithms reverse exponentials

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Read an exponential pair

Which exponent produces 1/4?



$2^{-2}=1/4$

Choose the output 1/4

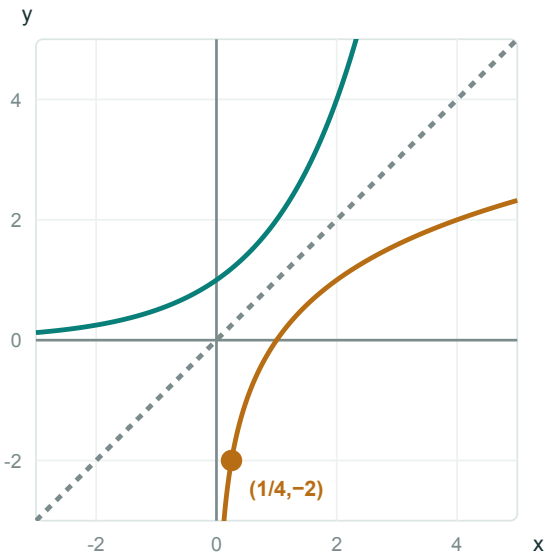
The exponent is −2.

Previous

Next →

Read the pair in reverse

Swap input and output coordinates.



$\log_2(1/4)=-2$

Show inverse relationship

A negative logarithm output is valid; its input is positive.

Previous

Next →

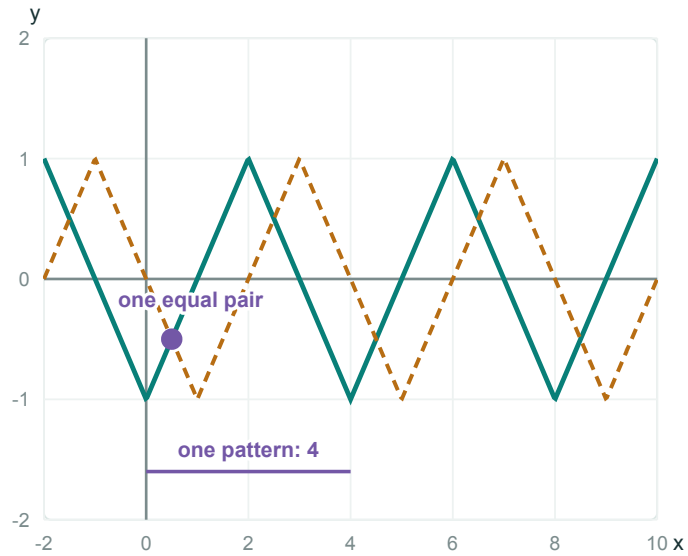
Equal axis scales preserve reflection in y=x. Teal: exponential; amber: logarithm.

FG-I1 / Periodicity beyond sine waves

ANNOTATED DESIGN SKETCH · selected states, not a production screen

One matching point is not enough

Slide the dashed copy one unit.



f(x) versus f(x−1)

Horizontal shift = 1

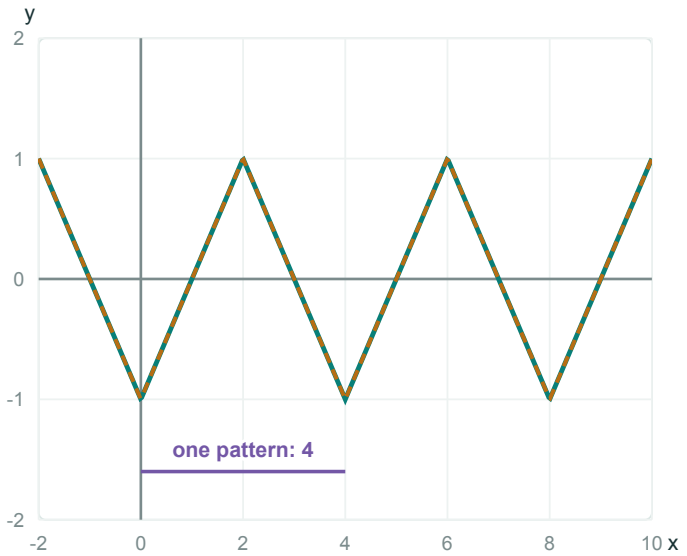
A pair of heights matches, but the whole pattern does not.

Previous

Next →

A whole pattern matches

Slide the copy four units.



f(x+4)=f(x) for all real x

Horizontal shift = 4

The smallest positive repeating shift is 4.

Previous

Next →

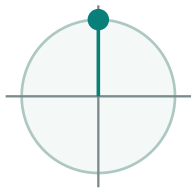
This example is defined to repeat over ℝ; a finite picture alone would not establish that claim.

FG-I2 / From a circle to sine and cosine

ANNOTATED DESIGN SKETCH · selected states, not a production screen

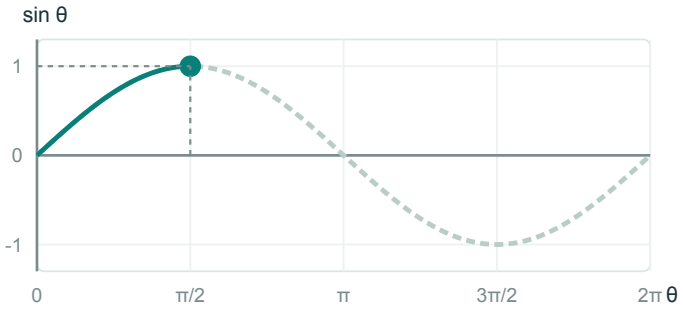
One quarter turn

Track the rotating point’s vertical coordinate.



θ = π/2

sin θ = 1



Circle height becomes the graph output

Angle θ = π/2

The input is an angle; the output is height 1.

Previous

Next →

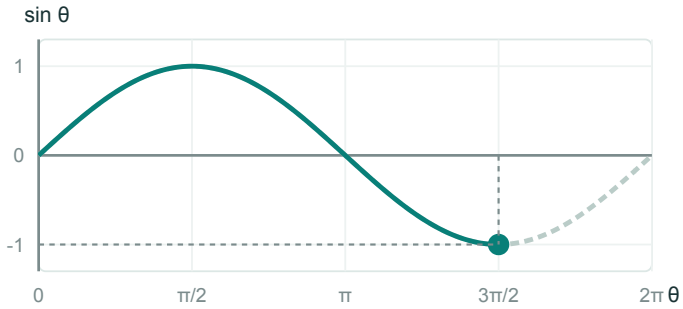
Three quarter turns

Follow the same coordinate below the axis.



θ = 3π/2

sin θ = −1



sin(3π/2)=−1

Angle θ = 3π/2

Signed circular position explains negative sine values.

Previous

Next →

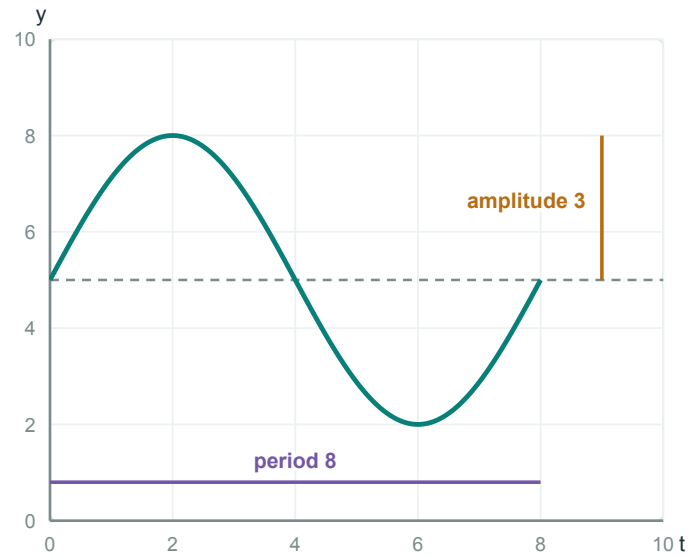
The graph horizontal axis is angle, not the circle point’s horizontal coordinate.

FG-I3 / Reading and building a wave

ANNOTATED DESIGN SKETCH · selected states, not a production screen

An eight-unit period

Identify midline, amplitude and period.



h(t)=5+3sin(πt/4)

Input coefficient B = π/4

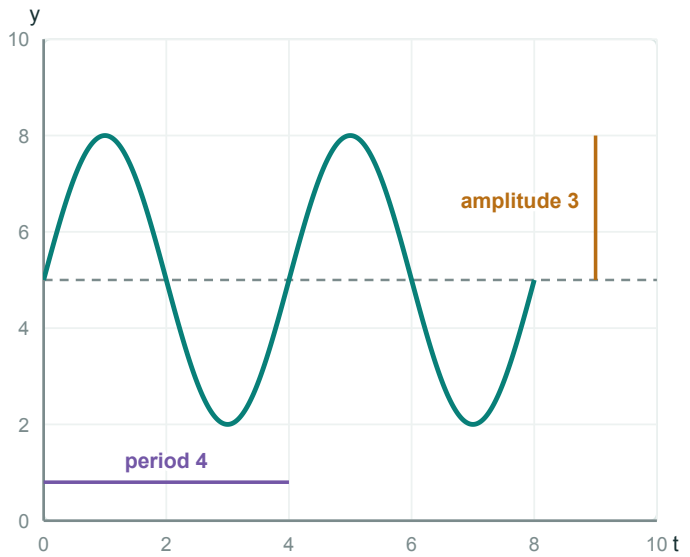
One cycle takes 8 time units.

Previous

Next →

A four-unit period

Double the input coefficient.



h(t)=5+3sin(πt/2)

Input coefficient B = π/2

The period halves; the range stays [2,8].

Previous

Next →

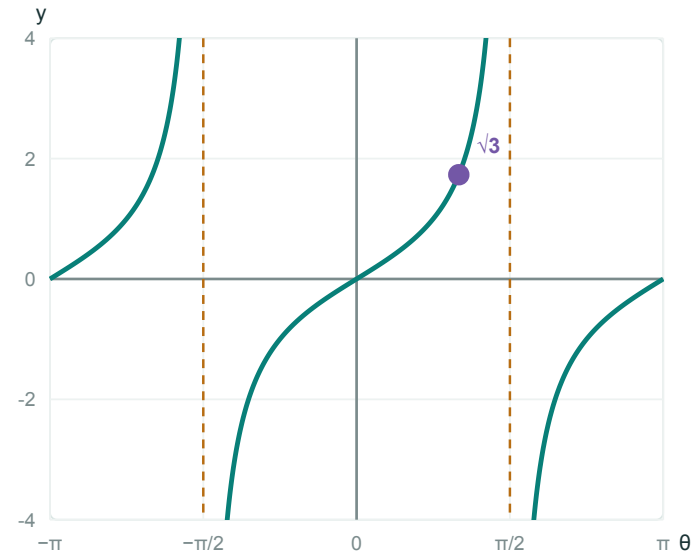
Midline, amplitude, period and phase have different controls and are introduced one at a time.

FG-I4 / Tangent: a periodic graph with gaps

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Before the excluded angle

Inspect sin θ divided by cos θ.



tan(π/3)=√3

Angle θ = π/3

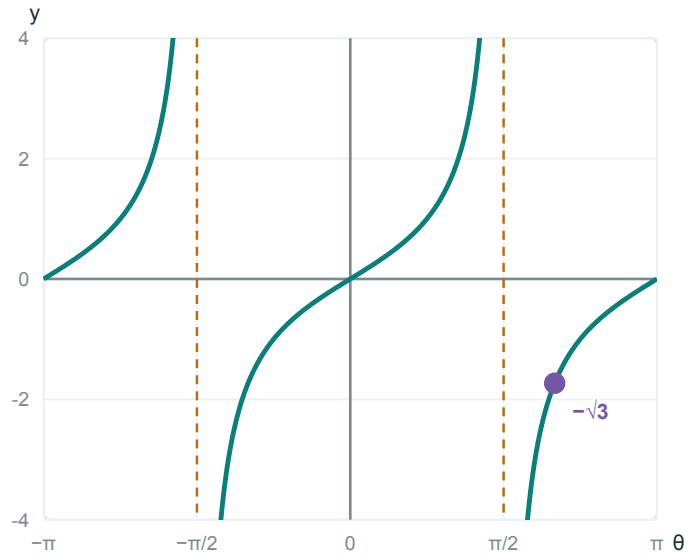
The ratio rises towards the gap at π/2.

Previous

Next →

After the excluded angle

Compare the denominator’s sign.



tan(2π/3)=−√3

Angle θ = 2π/3

At π/2 itself, the ratio is undefined.

Previous

Next →

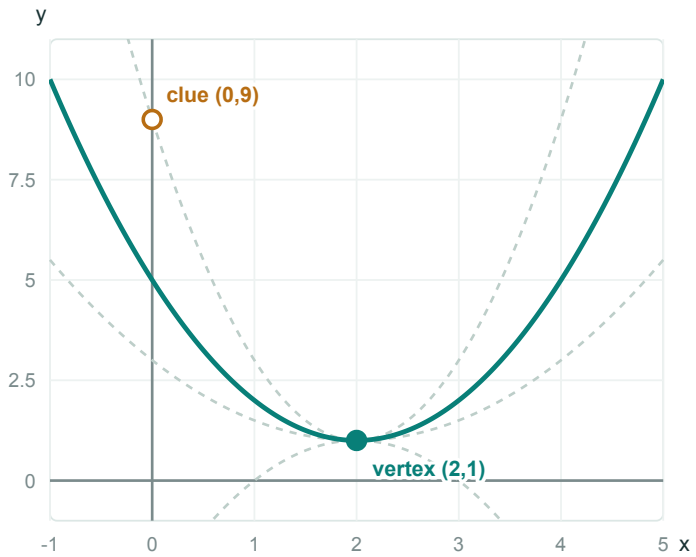
Branches are split at denominator zeros. A later stage compares copies separated by period π.

FG-J1 / Build a function from clues

ANNOTATED DESIGN SKETCH · selected states, not a production screen

The vertex leaves a family

Choose a curve with vertex (2,1).



f(x)=a(x−2)²+1

Remaining parameter a = 1

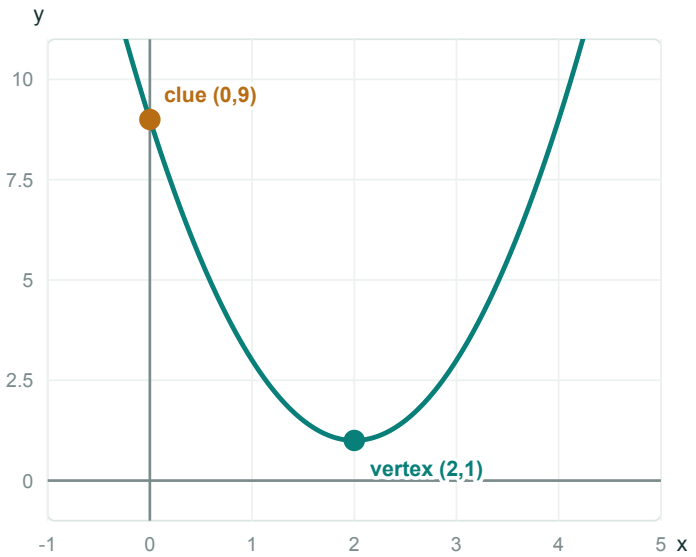
Many curves satisfy this first clue.

Previous

Next →

One more independent clue

Make the curve pass through (0,9).



9=4a+1 → a=2

Remaining parameter a = 2

Within the quadratic family, these clues fix the curve.

Previous

Next →

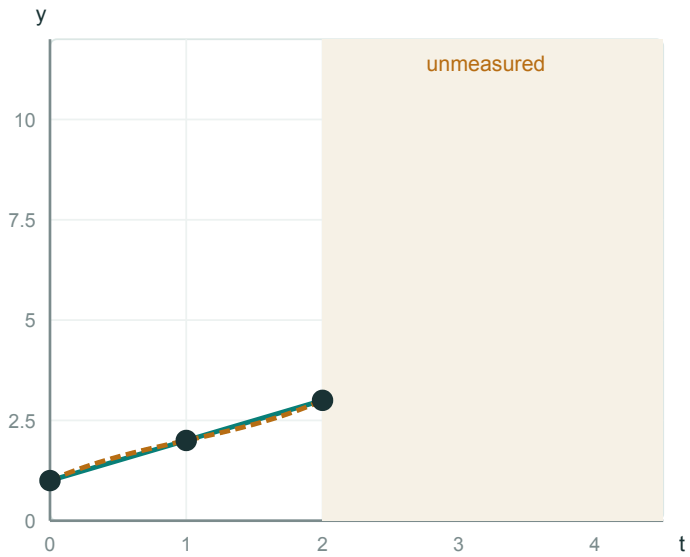
Ghost curves show alternatives; adding a redundant symmetric point does not add a new freedom.

FG-J3 / A perfect fit can still mislead

ANNOTATED DESIGN SKETCH · selected states, not a production screen

Three observed values

Check both models at t=0,1,2.



L=t+1; Q=L+¼t(t−1)(t−2)

Observed inputs: 0, 1, 2

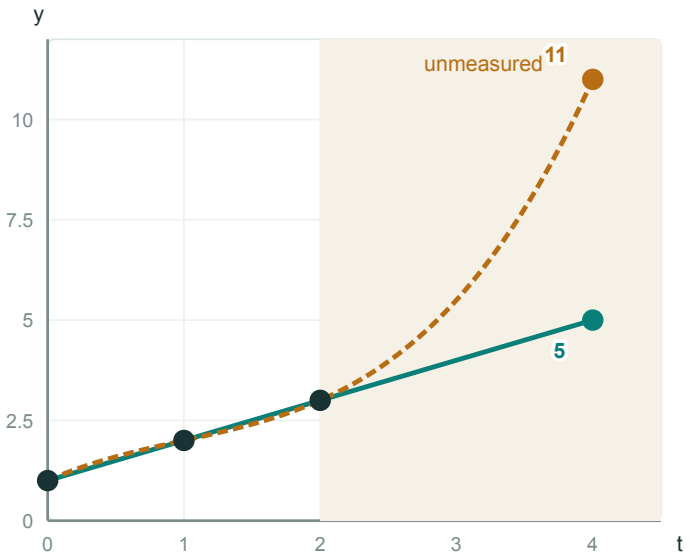
Both models fit every observation exactly.

Previous

Next →

Compare an unmeasured input

Reveal both predictions at t=4.



L(4)=5; Q(4)=11

Prediction input t = 4

Agreement with finite data does not force future agreement.

Previous

Next →

Fictional data. Solid: L; dashed: Q. Observation dots and prediction markers have distinct roles.